

ATTUEYI CODING ACADEMY

**GROUP 2
DATA ANALYSIS PROJECT**

TOPIC:
**CREDIT CARD FRAUD
DETECTION**

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0.1 EXECUTIVE SUMMARY

Owing to the rate of fraudulent acts perpetuated by our co-brothers in the tech-hacking departments, who have chosen to executive their franchise on the negative aspects, there is need to create the awareness for possible fraud activities so as to safeguard vulnerable victims who may happen to be our brethren.

This project seeks to analysis transactions made using credit cards in September 2013 for two days by the European Cardholders.

The dataset consists of 492 frauds detected out of 284,807 transactions.

It contains only numerical input variables and features V1, V2, ... V28 as principal components. It also contains the Time, Amount and Class features. The Time feature being the seconds elapsed between each transaction and the first transaction in the dataset. The Amount feature is the transaction amount and the Class feature is a response variable that takes value 1 in case of fraud and otherwise 0.

0.2 OBJECTIVES

The following are the objectives of this report

- i. To show that fraud activities were actually perpetuated.
- ii. To show the average amount siphoned through fraudulent acts.
- iii. To show the average possible time that transactions were made.
- iv. To use the class feature to decipher when transactions were fraudulent and otherwise non-fraudulent.

0.3 Data Sources

The Credit Card Fraud Detection Dataset used for the analysis was gotten online from Kaggle with reference to [Credit Card Fraud Detection](#).

0.4 METHODOLOGY

The tools deployed in analyzing the credit card fraud detection dataset were as follows;

- i. Microsoft Excel
- ii. SQL Server Management Studio (SSMS)
- iii. Power Business Intelligence (Power BI)
- iv. Microsoft Power Point
- v. Microsoft Word

The dataset was gotten as a Microsoft Excel Comma Separated Values Files (.csv), we carefully when through the gathered data to check for any error, inaccuracy and inconsistency in the data set for the purpose of cleaning the data in preparation for the analysis.

For the purpose of analyzing the data, we generated some relevant questions and we query the questions using SQL Server Management Studio (SSMS). Then we went on to using Power Business Intelligence (Power BI) to develop our visualization, while deploying Microsoft Power Point to create the background design for visualization. And finally, Microsoft Word was used to arrange the report.

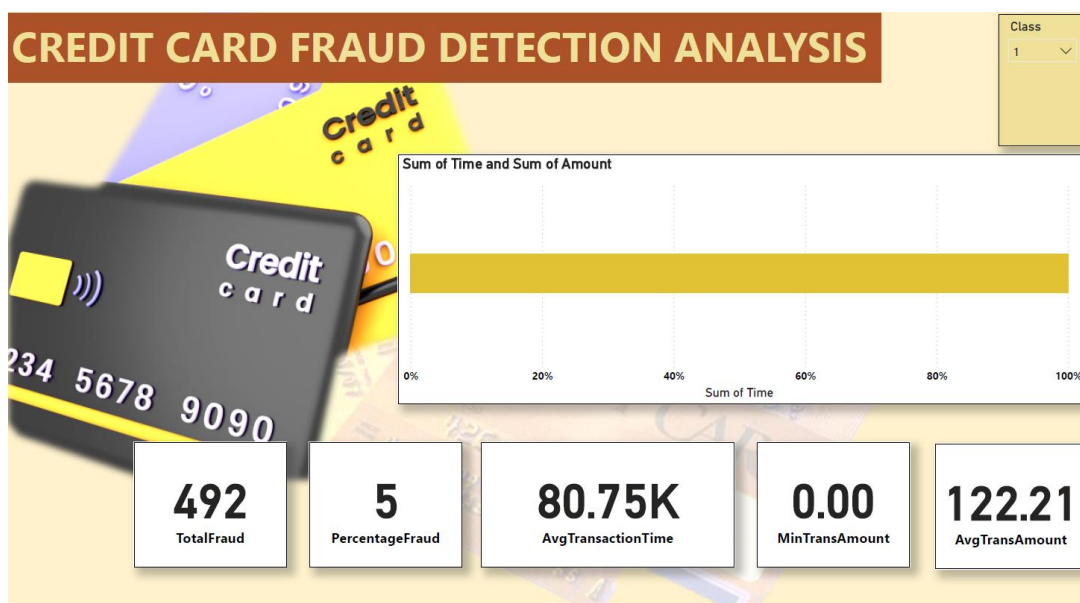
0.5 ANALYSIS AND KEY FINDINGS

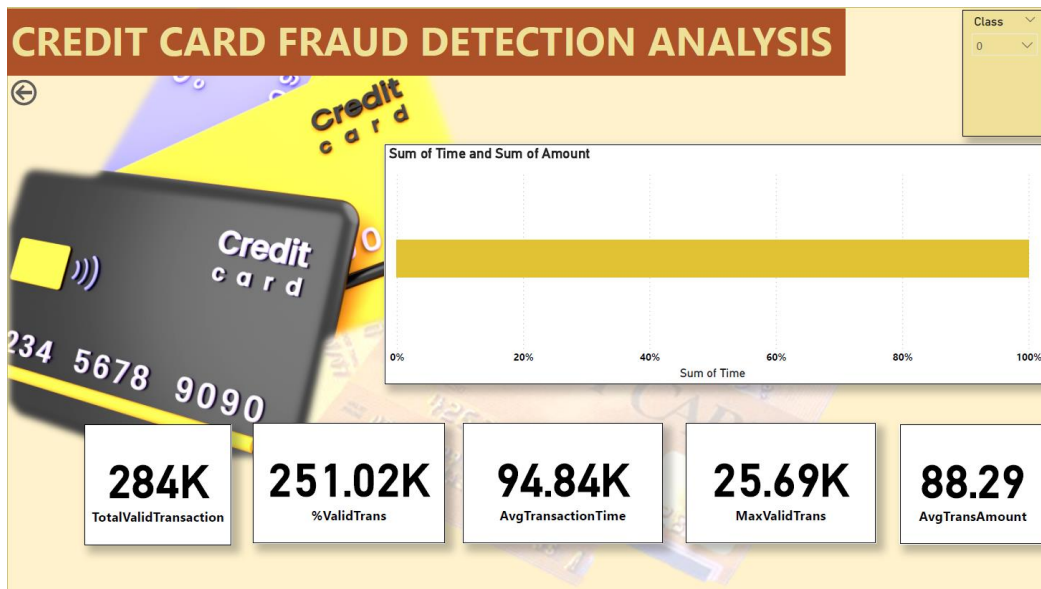
During the cleaning phase of the gathered data, we found out that the dataset did not contain any unique column that could be used for a primary key, as such it was ignored. We went further to create a database for the project and to import the credit card fraud detection data for further analysis but the data failed to import because the Time column originally came with the tinyint datatype which threw an error that hinders importing the file to SSMS, we changed the datatype to varchar(MAX) and the file importation was a success but failed when tried to query average transaction time. Then finally, we changed the Time datatype to Float and both importation and query was successful.

Also, the Amount datatype was initially a Float which posted a query challenge until we changed the datatype to Decimal and then the dataset was normalize for successful querying.

We went on to Importing the Credit Card Fraud Detection dataset to Power BI. There, we noticed that the dataset was loaded based on the first 200 rows. The dataset was imported as a txt/csv file. We checked to ensure that there were no errors found before loading the data for use. But after a successful loading of the data, we found error on the Amount column which showed different numbers which were not contained in the original dataset. So, we used Power Query to check for the error, only to realize that the data remained same as the original. We commenced working on our visualization while still observing the behave of the data whether it would be consistent with the original or not. Then we create measures to handle the calculations which all came out perfectly well, meaning the data weren't altered in the actual sense.

0.6 DATA VISUALIZATION





0.7 INTERPRETATION

Questions for querying the dataset:

- Determine the total number of fraudulent transactions and valid transactions in the dataset.
- What is the percentage of the class that was fraud?
- Determine the average time of the transactions.
- The least amount of transaction made in the dataset with a response variable(class) of 1.
- What is the average transaction amount in the dataset?

0.8 RECOMMENDATION

Based on the analysis of the credit card fraud detection dataset, we recommend that

- Sensitization program be organized for Credit Cardholders on the awareness of possible fraudulent activities.
- Banks should enhance their security measures in other to curb these fraudulent activities to save their customers.
- Credit Cardholders should report any fraudulent activities noticed.
- Banks should enable to send immediate messages to Cardholders so they can promptly alert when the transactions done is not from them.
- Banks should have a system that can reversed fraudulent transactions.

0.9 CONCLUSION

In conclusion, the analysis of the Credit Card Fraud Detection Dataset has brought to lamplight the possible fraudulent activities perpetuated on Credit Cardholders.

In view of the prevalent fraud activities, we note the following

- Creating awareness for Credit Cardholders on possible fraudulent activities.
- Enhancing security measures in the bank sectors.
- Immediate reporting of suspected or attempted fraud actions noticed for prompt restoration where possible.

We therefore proposed that every recommended phase to handle this fraud actions be taken into consideration in order to secure and protect Credit Cardholders and loss on the bank end.